

Do Medical MLLMs Use the Image? Controlled Tests of Image Dependence on Brain-MRI Question Answering

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Abstract

An accuracy drop when an image is removed or swapped is often read as evidence that a multimodal LLM uses the image, but it can also reflect response sensitivity alone: the answer changes without moving toward what the image supports. We ask when an image intervention measures correct visual tracking, using brain-MRI questions from the LUMIERE glioblastoma cohort whose answer keys have stated origins. Reading image ablation against the answer prior, and adding donor-image swaps scored against the donor’s key, separates accuracy benefit, output sensitivity and donor tracking. A reference model extracts sequence and lesion side from the image (95.7% and 100.0%, versus 21.7% and 36.8% without it) and tracks swapped images. Four open-weight models answer differently with the image but gain little from it: one image-effect test survives multiplicity correction, Llama-4-Scout on the diagnosis category, yet its accuracy with the image stays below the most common key’s share and its donor gain is not separable from a random donor assignment. On a stated management rule no open model exceeds the best category lookup. A motivating audit finds that a slice-based enhancement rule reproduces the expert RANO rating for 38% of the 197 follow-ups on which it applies, so we do not treat those ratings as validated targets. We characterize observable image-dependent behavior, not internal image use; keys are segmentation- or rule-derived and not clinician-validated.

1 Introduction

Medical multimodal LLMs are often evaluated by final-answer accuracy, which cannot distinguish image-based reasoning from reliance on textual cues, option patterns, or guessing. Such shortcuts are well documented: models exploit annotation artifacts in natural language inference (Gururangan et al., 2018), syntactic heuristics (McCoy et al., 2019), and unfaithful stated reasoning (Turpin et al.,

2023). In chest-radiograph question answering, models can rely on linguistic patterns or irrelevant image regions (Nguyen et al., 2025). Image ablation or swapping is informative only when the displayed image contains enough evidence to recover the answer.

We ask when an image intervention measures correct visual tracking rather than merely response sensitivity. An accuracy difference between an image arm and a text-only arm is confounded by the answer prior (a no-image arm that scores below chance inflates it), and an answer that changes when the image is swapped need not move toward the answer the new image supports. We therefore read the ablation together with three additions: reference values for each phase (chance and the most common key); a matched donor swap scored against the donor’s key with the no-image prior subtracted; and an audit of whether the key can be recovered from what is displayed, so that answerability is checked rather than assumed. Protocols that perturb the text or the image and report the change in accuracy, such as HEAL-MedVQA (Nguyen et al., 2025), address the same concern; we did not run their protocol, so the comparison is by design, not measured, and Section 5 gives a worked case in which the ablation alone points to image use and the additions do not. We instantiate the questions on the five OmniBrainBench phases (Peng et al., 2026) as same-patient chains from the LUMIERE glioblastoma cohort (Suter et al., 2022): anatomical and imaging assessment (AIA), lesion identification and localization (LIL), diagnostic synthesis and causal reasoning (DSCR), prognostic judgment and risk forecasting (PJRF), and therapeutic cycle management (TCM). The chain is an extension that lets us also test whether a model uses stated facts and upstream context in the management step. A motivating audit of the expert RANO ratings found that they cannot be treated as validated targets for the displayed slices

(Section 3.1), which motivates a benchmark with explicit key provenance and fixed controls (Sections 3.2–4).

The paper makes three contributions:

- A protocol that keeps three outcomes apart, accuracy benefit, output sensitivity and donor tracking, and reads image ablation against the answer prior, with a worked case (Section 5) in which the ablation alone credits Llama-4-Scout with image use on the diagnosis category although its accuracy with the image stays below the most common key’s share and its donor gain is not separable from a random donor assignment.
- An item set with an explicit origin for each key, built on LUMIERE, and a motivating audit of the expert labels: an enhancement-based rule reproduces the expert RANO category for 38% of the 197 follow-ups on which it is determinate (coverage is reported separately: it is undetermined for 44 of 241), so we do not treat those ratings as validated targets for the displayed slices; the audit does not show which side is wrong. The released OmniBrainBench groupings do not give same-patient chains (only 2 verifiable cross-phase links among 6,823 questions).
- An evaluation of four open-weight models and one reference model under a prespecified control suite. The reference model shows image-dependent performance on sequence and lesion-side questions, whereas the open models are limited and inconsistent; on the diagnosis-category task the most common key covers 58.7% of items, so accuracy there is hard to separate from a class prior even for the reference model. As an extension, the chain tests the use of stated facts and upstream context on the management step.

2 Related Work

Shortcuts and faithfulness. Models can reach correct labels without performing the intended reasoning. Hypothesis-only baselines exploit annotation artifacts in NLI datasets (Poliak et al., 2018; Gururangan et al., 2018), models adopt heuristics that fail on HANS (McCoy et al., 2019), and biased input features can alter chain-of-thought answers without appearing in the stated rationale (Turpin et al., 2023).

Answering without the image. Visual question-answering models often exploit language priors.

Goyal et al. (2017) countered this by pairing questions with images yielding different answers, while Chen et al. (2024) show that many multimodal benchmark items do not require visual content. Similar failures appear in medicine: Yan et al. (2025) report severe degradation under adversarial diagnostic counterparts, Jin et al. (2024) find flawed rationales in 35.5% of GPT-4V’s correct answers, and Nguyen et al. (2025) (HEAL-MedVQA) use textual and visual perturbations to localize shortcuts in chest-radiograph QA. We extend this logic to brain-MRI questions, and add to image-only controls a donor swap scored against the donor’s key with the no-image prior subtracted, reference values for each phase, and an audit of whether the key is recoverable from the displayed inputs; we did not run HEAL-MedVQA’s protocol, so the difference is one of design. The same-patient chain extends the questions to interventions on supplied clinical facts. UniMod (Gu et al., 2026) addresses modality imbalance during training by supervising image-only, text-only, and fused predictions separately; we instead evaluate existing models for observable image dependence.

Brain benchmarks and longitudinal data. OmniBrainBench (Peng et al., 2026) contains 6,823 closed-ended brain-imaging questions spanning five clinical phases and 15 modalities. Its supplement describes a patient-level ADNI extension, but we did not evaluate it. LUMIERE (Suter et al., 2022) and MU-Glioma-Post (Mahmoud et al., 2025) provide longitudinal glioma imaging, treatment, and outcome data but no question layer.

3 Building Chains from LUMIERE

Why not OmniBrainBench. A chain needs one patient’s phases. We checked whether any image is shared across questions of different phases within a grouped case: of 6,823 closed-ended questions only 2 verifiable cross-phase patient links exist (both in VQA_RAD, one AIA and one LIL question each), and a second method based on filename identifiers agreed; the counts are lower bounds (Appendix A). Each sub-corpus was built for one task, so a patient is essentially asked one phase-type of question. This is an incompatibility between the released groupings and our evaluation, not a defect in a promise its authors made.

LUMIERE. LUMIERE (Suter et al., 2022) has 91 glioblastoma patients with longitudinal MRI

(T1, contrast-enhanced T1, T2, FLAIR), automated segmentations (DeepBraTumIA and HD-GLIO-AUTO; Suter et al., 2022; Kickingereder et al., 2019), RANO ratings by one neuroradiologist (Wen et al., 2010), and overall survival for most patients. Segmentations are automated and were not manually verified; the ratings are expert-assigned.

3.1 Motivating audit: can an automated rule reproduce the expert ratings from the displayed slices?

The natural key for a diagnosis-category question is the expert RANO rating, but whether that rating can be recovered from the displayed slices is untested. We tested it with an automated reproducibility audit: for each first-line follow-up we tracked one enhancing lesion across scans (at each scan, the 3D-connected enhancing component nearest to the largest component of the post-operative baseline), measured it on its own largest axial cross-section, computed the RANO bidimensional product (longest diameter times longest perpendicular diameter) and applied the enhancement criteria (progression: measurable disease at least 25% above the smallest earlier measurement, the nadir, or measurable disease where the nadir had none; response against the post-operative baseline; measurable means both diameters at least 10 mm). The rule has no new-lesion criterion beyond that. Coverage: of 241 first-line follow-ups (one per rated, imaged timepoint before any second surgery), 44 are undetermined because no measurable enhancing disease exists on the baseline or nadir scan, so the rule abstains on them. Agreement, our primary number, is computed over the 197 determinate follow-ups: the rule reproduces the expert category for 75 (38%, Wilson 95% interval 32 to 45%, and 29 to 48% when the 59 patients (up to 15 follow-ups each) are resampled instead; counting the undetermined ones as failures would give 31%, which mixes abstention with disagreement): progressive disease 59 of 146, complete response 13 of 15, stable disease 2 of 65, partial response 1 of 15 (these denominators include the undetermined ones). Of the 122 disagreements, the rule returns complete response for 90 (the expert rated progressive disease for 46 and stable disease for 33) and progressive disease for 16 (13 rated stable by the expert). Disagreement can arise from errors in the automated segmentation or in our slice choice and measurement, from a different target lesion or

reference scan than the rater's, from differences between our simplified rule and the rater's assessment (which may use T2/FLAIR change, clinical status and steroid dose), or from evidence absent from the displayed slices; this audit does not separate them. LUMIERE documents that one neuroradiologist rated the studies but not what information they used. Our automated product correlates only moderately with the diameters the rater recorded for the target lesion (Spearman 0.39 over 81 follow-ups; ours is a median 1.3 times larger), so measurement error is one plausible contributor, although agreement did not differ with size closeness in the subset that could be checked (Appendix D). The audit therefore shows that these ratings cannot be treated as validated targets for the displayed inputs; it does not measure how many of them are unanswerable, which would need qualified reviewers to assess the displayed inputs for a sample of the disagreements.

3.2 The item set

The keys have four different origins (Table 1), and we make no uniform claim that each is a function of what the model sees. AIA: the sequence rendered (acquisition-derived). LIL and DSCR: derived from automated segmentation masks that the model does not see, so a reader can recover them from the slices only to the extent that the segmentation is right and the displayed slice shows it; they are segmentation-derived targets that reviewers have not yet validated. PJRF: the recorded outcome (death within 52 weeks), scored probabilistically, since a future outcome is not a function of the facts stated. TCM: the output of an explicitly stated management rule applied to the category and timing given in the stem, which tests following a stipulated rule, not clinical correctness. All random choices use fixed seeds and never look at model output.

Cohort. Of the patients with a first-line follow-up (rated, imaged, before any second surgery, with an imaged post-operative reference scan), we select one follow-up per patient in strata defined by the TCM rule class (continue, repeat MRI first, change therapy) and by the DSCR category, with quotas fixed in advance and a fixed seed, giving 46 patients (counts by phase and key in Appendix D). Selection used tabular records and segmentation masks only. Progressive-disease follow-ups whose timing falls within 2 weeks of the 12-week window are left out.

Phase	Question	Model-visible inputs	Key	<i>n</i>
AIA	Which MRI sequence is shown?	one axial slice (T1, T1 after contrast, T2 or FLAIR)	the sequence rendered (balanced)	46
LIL	Hemisphere and anterior/posterior half of the main tumor region	follow-up contrast-enhanced T1 slice at the level of the largest tumor region	position of the largest tumor-core component on that slice, relative to the midline and the brain's front-to-back midpoint	38
DSCR	RANO enhancement category at follow-up	baseline, follow-up and (when different from the baseline) nadir slices at the largest cross-section of the tracked enhancing lesion; the RANO thresholds are stated	category from the bidimensional products on those slices	46
PJRF	Probability of death within 52 weeks of this scan	age, sex, MGMT, IDH, extent of resection, weeks since surgery; alive at the scan	recorded outcome; scored by Brier score, not accuracy	45
TCM	Next management step	the same clinical facts plus weeks since chemoradiotherapy; the category comes from the chain	rule: non-PD continue; PD within 12 weeks of radiotherapy repeat MRI first; PD later change therapy	46

Table 1: The phases. Stems for AIA, LIL and DSCR contain no patient-specific fact and every option set contains every possible answer, so a swapped image never conflicts with the text and the donor’s answer is always available. DSCR shows two images, or three when the nadir differs from the baseline.

Images. Each image is an atlas-space axial slice, skull-stripped by the dataset authors with HD-BET (Isensee et al., 2019), at 4× magnification with a 20 mm scale bar, in radiological orientation (checked against the affine). Slice levels are those on which the key is computed.

AIA, LIL, DSCR. AIA is the positive control: its key is the sequence rendered, balanced over four sequences (the most common covers 26.1% of items), so text alone can reach only chance in expectation. LIL’s key is the hemisphere and anterior/posterior half of the largest tumor-core component on the displayed slice; lesions within 10 mm of a midline, smaller than 100 mm² or with under 80% of their pixels on their centroid’s side of the midline are not asked about. DSCR’s stem states the RANO enhancement criteria, and its key is the category computed from the bidimensional products. The expert rating is recorded for each item but is not the key.

PJRF. The prediction time is the follow-up scan and the outcome is death within 52 weeks of it, from the LUMIERE survival record. The model picks one of four probability bins (midpoints .10, .30, .50, .80) and is scored with the Brier score, not by whether it picked a “correct” bin. The recorded outcome is death in 62.2% of 45 patients, a base rate that reflects our selection and is not a population rate. The origin and censoring of the survival time are undocumented (Appendix F), so PJRF is secondary and no other claim in this paper rests on it.

TCM. LUMIERE records no treatment after chemoradiotherapy, so a documented treatment is not available and the key is a rule-based recommendation, not what was done: non-progressive disease continues adjuvant temozolomide; progressive disease within 12 weeks of completing radiotherapy is confirmed with repeat MRI before any change; later progressive disease prompts a change of therapy. The 12-week window is from the RANO criteria (Wen et al., 2010), which also allow progression inside the window when new enhancement lies outside the radiation field; we cannot assess the field, so the rule is a simplification. Radiotherapy dates are not in LUMIERE, so the stem states the weeks since chemoradiotherapy under an assumed standard schedule (ending 10 weeks after surgery). The same diagnosis therefore maps to different actions depending on the stated timing (7 repeat-MRI and 20 change-therapy items among 27 progressive-disease items), which lets a lookup of the category alone be told apart from use of the timing fact.

4 Controls and Pre-specified Analysis

The analysis plan (hypotheses, margin, correction family and decision rules) was written before any model run; its hash is recorded in the project log, and the file is not externally timestamped. Models are the four open-weight models (MedGemma-4B (Søllergren et al., 2025), Gemma-3-12B and Gemma-3-27B (Gemma Team, 2025), Llama-4-Scout; 4-bit, greedy decoding, 800 tokens; unparseable responses score incorrect). Each con-

dition asks one question with a manufactured upstream context, so a phase’s effect is not mixed with the model’s own earlier answers.

- **E1, image effect** (AIA, LIL, DSCR): accuracy with the image minus text only, paired by patient, with a patient-bootstrap interval. The margin is ± 10 percentage points, fixed in advance; “within margin” requires the whole interval inside it, and the interval half-width is about $1.96\sqrt{d/n}$ for a share d of items whose correctness differs, so at $n = 46$ the margin is reachable only if $d \leq 12\%$ (10% at $n = 38$). Holm correction over the 12 model-phase tests; other contrasts are descriptive and unadjusted, with exact McNemar tests (McNemar, 1947) and Wilson intervals (Wilson, 1927) where a proportion is reported. Because the bootstrap collapses to zero width when every pair is concordant, we also report, post hoc, an add-half interval for the paired difference (Agresti and Min, 2005) and the exact p . **AIA is the positive control:** the plan’s operational reading rule is that for a model whose unadjusted AIA interval is not above zero, the test has not been shown to detect image use, and E1 on LIL and DSCR supports no conclusion about it.

- **E2, donor tracking:** swap the image for a donor’s with a different key (one donor per patient and phase, same number of images); $G = 1[\text{swap} = \text{donor key}] - 1[\text{text only} = \text{donor key}]$, so the no-image prior is subtracted. The estimand is conditional on the fixed assignment, in which a patient donates to at most two recipients; donor-dependence checks are post hoc (Appendix D).

- **E3, use of stated facts** (TCM): with the true context, move the stated timing across the 12-week window (*flip_window*) or replace the stated category by one that maps to another action class (*flip_label*). The registered summary was the share of answers equal to the rule’s new answer; because that counts answers that had not changed, we report each item’s transition relative to the same item under the true context (moved to the rule’s new answer, already there, unchanged elsewhere, changed to another answer), and take moved over items whose original answer was not already the new one as the rate of interest. A category lookup alone reaches at most 84.8% accuracy (56.5% if it always confirms first) and cannot follow the timing flip.

- **E4, upstream context:** true, randomly wrong (each earlier answer replaced by a seeded random incorrect option) or absent context.

- **E5, PJRF** (secondary; results in Appendix F): Brier score and skill against the leave-one-out base rate, and AUC.

We report three observable outcomes separately: accuracy benefit (E1), output sensitivity (the share of items whose answer letter differs between the image and text-only arms, and between the swapped and own image), and correct donor tracking (E2). None of them observes internal processing: a model can use an image and still answer wrongly, and an unchanged answer does not show unchanged processing. Output sensitivity, the transition tables and the output-handling counts are post hoc, descriptive analyses. The ordinary five-phase chain (own answers as context) is run with and without images as a secondary analysis; each arm regenerates its upstream answers, so later-phase differences are total effects through the chain.

Reference model. Three of the four open models did not meet the operational AIA gate (Section 5), so we added one reference (positive-control) model to the plan post hoc: Gemini-3.6-Flash (gemini-3.6-flash, Google API). It runs E1 and E2 only, with the same prompts, temperature 0 and decision rules. The API forces a 16,384-token output cap, because hidden reasoning tokens count against it. The control is reported apart from the four-model Holm family, empty API responses are counted and rerun (there were 0 among 386 records), and, being an API model, it can change without notice. A second post hoc addition is a precision control: Gemma-3-27B is rerun in bf16 (Google’s gemma-3-27b-it weights, vLLM, one GPU) on the same three conditions and phases, reported apart from the Holm family. Its reading rule was fixed in advance: an AIA interval above zero and clearly above the 4-bit model’s would mean quantization explains part of the gap, and a result near chance that it does not; we do not adopt the second reading (Results). The serving stack differs from the Ollama runs, so precision and stack are confounded.

5 Results

5.1 Reference values without a model

Table 2 gives what constants and lookups reach. AIA is close to uniform over four answers, DSCR’s

Phase	n	Chance	Most common key
AIA	46	25.0	26.1%
LIL	38	25.0	28.9%
DSCR	46	25.0	58.7%
TCM	46	25.0	43.5% (lookup $\leq$ 84.8%)
PJRF	45	(Brier)	base rate 62.2%

Table 2: Non-model references on the items: chance, share of the most common key, the ceiling of a category-only TCM lookup, and the PJRF base rate.

most common category is progressive disease for 58.7% of items, and a category-only TCM lookup reaches at most 84.8%.

5.2 Image controls and stated facts

The reference model reads the image. Gemini-3.6-Flash identifies the sequence for 95.7% of patients with the image and 21.7% without it (difference +73.9 points, 95% CI [+58.7, +87.0]), and it places the lesion for 100.0% against 36.8%. With the image swapped for a donor’s it gives the donor’s answer for 95.7% (AIA) and 100.0% (LIL) of patients (Table 7). A model that reads the slices can therefore answer these items, which shows the tasks are feasible; it does not show that every item is valid, and a weak result for another model on them reflects that model’s performance on the task, whether from perception, input handling or training. It misses only 2 AIA items and 0 LIL items. DSCR is different. The reference model scores 71.7% with the image and 58.7% without (Δ +13.0 [+0.0, +26.1]), where the most common key covers 58.7% of items, so most of its text-only accuracy is the class prior. Its answers do depend on the image, since a swap changes 57.1% of them (gain +16.7 [+2.4, +31.0]), but even this model recovers the key for only 71.7% of items, so the DSCR items are only partly answerable from the slices, whether from perception, from segmentation-derived keys the model does not see, or from key error, which we cannot separate. A null DSCR effect for another model therefore cannot be read as a failure to use the image.

Open models: image effect (E1). Sequence identification is 26.1 to 37.0% (chance 25%; Table 3). One E1 test survives Holm correction, Llama-4-Scout on DSCR (+45.7 [+28.3, +63.0], adjusted $p = 0.00$); every other adjusted p is at least 0.69. That gain does not show that Scout reads the category from the slices: with the image its accuracy (54.3%) stays below the share of the

most common key (58.7%), and the gain arises because without the image it falls to 8.7%, far below that share. The plan’s operational AIA gate (unadjusted interval above zero) is met only by Llama-4-Scout (+13.0 [+2.2, +26.1]), whose gain is small beside the reference model’s, and not by MedGemma-4B (+8.7 [−2.2, +21.7]), Gemma-3-12B (+0.0 [−13.0, +13.0]) or Gemma-3-27B (+6.5 [−2.2, +17.4]); the gate depends on interval width. For the three models below the gate, the plan’s reading rule permits no interpretation of their LIL or DSCR effects. On LIL every interval includes zero, with point estimates from +5.3 to +7.9; on DSCR Gemma-3-27B is within the margin (+0.0), while MedGemma-4B (+8.7 [+2.2, +17.4]; exact $p = .125$, and the add-half interval [−0.5, +17.2] includes zero) and Gemma-3-12B (+10.9 [+2.2, +19.6]) have unadjusted intervals above zero (adjusted $p = 1.00$ and 0.69) that the reading rule does not let us interpret; MedGemma-4B’s accuracy with the image (19.6%) is below chance and Gemma-3-12B’s (58.7%) equals the most common key’s share. A test that does not survive correction is not evidence of no effect: every open-model AIA and LIL interval is wide and none lies inside the margin. Each open model’s AIA answers also concentrate on one sequence (its most frequent answer covers 52 to 76% of the 46 items), which is what a fixed default plus weak image use would look like; we cannot separate that from a failure to perceive the sequence, although every image does reach the model in each pipeline (Appendix F). Accuracy benefit is small, but answers are not insensitive to the image: the answer differs between the image and text-only arms for 26 to 33% of AIA items, 18 to 47% of LIL items and 0 to 88% of DSCR items across the four models (Table 3).

Precision control. Rerun in bf16, Gemma-3-27B identifies the sequence for 26.1% of patients with the image and 26.1% without it (Δ +0.0 [−10.9, +10.9]), against 32.6% and 26.1% at 4-bit and 95.7% for the reference model. The interval includes zero and the accuracy stays near chance (25%), so the rerun did not restore AIA performance, and this experiment gives no evidence that changing precision and serving stack resolves the gap. Precision and stack changed together, and a null result cannot show that quantization has no effect, so we draw no conclusion about quantization. On LIL and DSCR the bf16 model is also indistinguishable from text-only (+0.0 and +0.0),

Model	Phase	<i>n</i>	Image	Text-only	Δ [95% CI]	Add-half CI	Exact <i>p</i>	Answers differ	Interval
MedGemma-4B	AIA	46	34.8	26.1	+8.7 [−2.2, +21.7]	[−3.7, +20.4]	.289	30.4	inconclusive
MedGemma-4B	LIL	38	23.7	18.4	+5.3 [−5.3, +15.8]	[−5.8, +15.8]	.625	18.4	inconclusive
MedGemma-4B	DSCR	46	19.6	10.9	+8.7 [+2.2, +17.4]	[−0.5, +17.2]	.125	19.0	CI above 0
Gemma-3-12B	AIA	46	26.1	26.1	+0.0 [−13.0, +13.0]	[−12.2, +12.2]	1.00	32.6	inconclusive
Gemma-3-12B	LIL	38	31.6	23.7	+7.9 [−7.9, +23.7]	[−9.3, +24.3]	.549	47.4	inconclusive
Gemma-3-12B	DSCR	46	58.7	47.8	+10.9 [+2.2, +19.6]	[+0.9, +20.0]	.062	11.9	CI above 0
Gemma-3-27B	AIA	46	32.6	26.1	+6.5 [−2.2, +17.4]	[−3.6, +16.1]	.375	26.1	inconclusive
Gemma-3-27B	LIL	38	23.7	18.4	+5.3 [−5.3, +15.8]	[−5.8, +15.8]	.625	34.2	inconclusive
Gemma-3-27B	DSCR	46	58.7	58.7	+0.0 [+0.0, +0.0]	[−4.1, +4.1]	1.00	0.0	within ±10 pp
Llama-4-Scout	AIA	46	37.0	23.9	+13.0 [+2.2, +26.1]	[+0.8, +24.2]	.070	30.4	CI above 0
Llama-4-Scout	LIL	38	39.5	34.2	+5.3 [−10.5, +21.1]	[−11.2, +21.2]	.754	44.7	inconclusive
Llama-4-Scout	DSCR	46	54.3	8.7	+45.7 [+28.3, +63.0]	[+26.0, +61.5]	<.001	88.1	CI above 0

Table 3: E1: accuracy (%) with the image and with text only, with no upstream context, the paired difference in percentage points with its bootstrap interval and an add-half sparse-data interval (Add-half CI), the exact McNemar *p* (unadjusted), the share of items whose answer letter differs between the two arms (answers differ), and the interval reading (unadjusted; only Llama-4-Scout on DSCR survives Holm correction, and “CI above 0” is not otherwise a confirmatory result). AIA is the positive control.

	TCM acc. (%)			Moved / informative	
	true	wrong	none	label (all)	timing (PD)
MedGemma-4B	54.3	39.1	34.8	10/25	1/27
Gemma-3-12B	69.6	30.4	39.1	18/40	8/21
Gemma-3-27B	69.6	32.6	45.7	25/46	0/20
Llama-4-Scout	54.3	37.0	45.7	10/37	11/21

Table 4: E3: TCM accuracy (%) under true, random wrong and absent upstream context, and the number of answers that move to the rule’s new answer after the stated category is replaced (label, all items) or the stated timing crosses the 12-week window (timing, progressive-disease items), over the informative items, those whose true-context answer was not already the new answer.

and donor tracking on AIA is +0.0 [−8.7, +8.7]. Only one model was rerun.

Open models: donor tracking (E2). Swapping the image changes the answer for 33 to 50% of AIA items and 13 to 55% of LIL items, but the new answer follows the donor’s key with an interval above zero in only two cells (Table 6): MedGemma-4B on AIA (gain +10.9 [+2.2, +19.6]; exact *p* = .062, add-half interval [+0.9, +20.0]) and Llama-4-Scout on DSCR (+28.6 [+7.1, +50.0]; exact *p* = .029; the two exact *p* become .688 and .348 under an exploratory Holm correction over the twelve E2 cells, outside the registered family; a swap changes 47.6% of its DSCR answers), the phase in which its accuracy gain also survives correction. Both survive resampling by donor and dropping any one donor, but neither is separable from a random assignment of donor keys (permutation *p* = .059 and .244; Appendix D), so for DSCR we cannot tell donor-specific tracking from a shift that any swapped image would produce. Every other in-

terval includes zero, including Llama-4-Scout and Gemma-3-27B on sequence and all four models on lesion side, where the MedGemma-4B gain is negative (−7.9 [−18.4, +2.6]). A swap changes 7.1% of Gemma-3-27B’s DSCR answers. On sequence the MedGemma-4B gain is far below the reference model’s (+67.4), as are all open-model lesion-side gains (reference +86.8).

Worked example: ablation without tracking.

Llama-4-Scout on DSCR is the case in which reading the ablation alone would credit image use. Its image-minus-text difference is the largest and the only one that survives correction (+45.7 [+28.3, +63.0], adjusted *p* < .01; 24 items correct only with the image, 3 only without). Three further readings, each added by this protocol, change the conclusion. First, its accuracy with the image (54.3%) stays below the share of the most common key (58.7%), so the image arm does not beat a constant answer. Second, its text-only accuracy is 8.7%, far below both the 25% chance level and the most common key’s share, so the difference arises chiefly because the no-image arm scores far below the prior, not because the image arm exceeds it. Third, when the image is swapped for a donor’s, its answer follows the donor’s key with a gain of +28.6 [+7.1, +50.0] over the text-only prior, but a permutation of the donor assignment does not separate that gain from a random assignment (*p* = .244). The ablation therefore shows response sensitivity to the image, not that the category is read from the slices; the additions are what prevent that reading.

Stated facts and context (E3, E4). With the true context TCM accuracy is 54.3, 69.6, 69.6 and 54.3% for MedGemma-4B, Gemma-3-12B,

Gemma-3-27B and Llama-4-Scout, all below the best category-only lookup (84.8%, which escalates every progressive-disease item), although Gemma-3-12B and Gemma-3-27B exceed the confirm-first lookup (56.5%). That is a comparison of accuracy; it does not show which facts a model uses, so we compare each item with itself under the true context (Tables 4 and 9). When the stated category is replaced, the answer moves to the rule's new answer for 10 of 25 informative items for MedGemma-4B, 18 of 40 for Gemma-3-12B, 25 of 46 for Gemma-3-27B and 10 of 37 for Llama-4-Scout; for MedGemma-4B, 21 of 46 answers already equalled the new rule answer before the flip because it answers the most common class almost regardless (continue items 100% correct, the other two classes 29% and 20%), so counting them as following the rule would overstate its use of the label. When the stated timing crosses the 12-week window on progressive-disease items, the answer moves to the rule's new answer for 1 of 27, 8 of 21, 0 of 20 and 11 of 21 informative items (same model order): Llama-4-Scout changes its answer with the stated timing most often, Gemma-3-12B less often, MedGemma-4B rarely and Gemma-3-27B never (its 7 answers equal to the rule's new answer were already there). True context beats no context by +19.6 [+6.5, +32.6] points for MedGemma-4B, +30.4 [+17.4, +43.5] for Gemma-3-12B and +23.9 [+10.9, +37.0] for Gemma-3-27B, and beats random wrong context on TCM in all four models (by +15.2, +39.1, +37.0 and +17.4 points in the model order above). For LIL and DSCR the true context does not help in any model; the one interval that excludes zero is a loss, for Gemma-3-12B on LIL against no context. Stating the management rule in the prompt (post hoc, descriptive) raises TCM accuracy over the true context alone for Gemma-3-12B, Gemma-3-27B and Llama-4-Scout and leaves MedGemma-4B unchanged: MedGemma-4B 54.3 to 54.3% (+0.0 [−6.5, +6.5]), Gemma-3-12B 69.6 to 82.6% (+13.0 [+0.0, +26.1]), Gemma-3-27B 69.6 to 80.4% (+10.9 [+2.2, +19.6]) and Llama-4-Scout 54.3 to 67.4% (+13.0 [+4.3, +23.9]). MedGemma-4B and Llama-4-Scout answers were invalid JSON for 16 and 8 of the 46 items under the true context and for 3 and 0 with the rule stated, so part of Scout's gain may be better-formed output. With the rule stated, Gemma-3-12B and Gemma-3-27B reach 82.6% and 80.4%, close to the best category lookup (84.8%) but not above it.

Chain (secondary). With the model's own answers as context, most image-minus-text differences are small and their intervals include zero. The exception is Llama-4-Scout on AIA (+13.0 [+2.2, +23.9]), which repeats its E1 result. Its DSCR difference (+17.4 [+0.0, +32.6]) has a lower limit at zero and is much smaller than its E1 gain. The largest other difference is Gemma-3-12B on LIL (+15.8 [−5.3, +34.2]). These are not corrected for multiplicity.

What the chain adds. A chain can show whether a model uses its own earlier answers or the stated diagnosis in a later step. Here that matters for TCM only: true upstream context beats no context in MedGemma-4B, Gemma-3-12B and Gemma-3-27B (unadjusted), and the fact flips show which stated facts move the answer (Tables 4 and 9), whereas for LIL and DSCR the true context does not help and the ordinary chain gives small total effects. We do not claim a causal-degradation effect across phases.

What this shows and does not show. We separate three observable outcomes. *Accuracy benefit:* the reference model gains +73.9 and +63.2 points from the image on sequence and lesion side, whereas the open models gain +0.0 to +13.0 points on sequence with an interval that excludes zero only for Llama-4-Scout, not surviving Holm correction; the one open-model result that survives it, Llama-4-Scout on DSCR, stays below the most common key's share. *Output sensitivity:* answers do change with the image in all four open models, so the image is not simply ignored (Table 3). *Donor tracking:* swapped-image answers move toward the donor's key with an interval above zero only for MedGemma-4B on sequence (+10.9 points) and Llama-4-Scout on the diagnosis category (+28.6), against +67.4 and +86.8 for the reference model on sequence and lesion side; a permutation of the donor assignment does not separate either open-model result from a random donor (Results). The evaluated open models therefore show limited and inconsistent image-dependent performance on these tasks, and the reference model shows strong image dependence on items it can answer. This does not identify a mechanism: a model can use an image and still answer wrongly, and unchanged answers do not show unchanged processing, so we claim neither non-use nor learned shortcuts; model size and training differ from the reference model and could produce the same gap.

On the management rule no model exceeds the best category lookup in accuracy, stating the rule in the prompt raises accuracy for three of the four, and their sensitivity to the stated category and timing differs by model (Tables 4 and 9). For DSCR the most common key covers 58.7% of items, and Gemma-3-27B (both arms) and Gemma-3-12B (with the image) score exactly that, so accuracy on DSCR alone cannot separate image use from that prior, and we draw no conclusion about image use from it. Input handling was checked only in part (Appendix F), and none of this is clinician-validated.

6 Conclusion

Accuracy on an expert-labeled brain-MRI chain cannot show whether models read the scan when the labels are not validated targets for what is displayed: an automated rule reproduces the expert RANO rating for 38% of the follow-ups on which it is determinate, a discrepancy we cannot yet attribute; the benchmark we started from did not establish same-patient chains. We build the chain on LUMIERE with a stated origin for each key and fixed a control suite in advance, with a positive control that only the image can answer, matched donor swaps, fixed-context comparisons and fact-flip tests of a management rule. A reference model reads the sequence and the lesion side from the image (95.7% and 100.0%, against 21.7% and 36.8% without it). Four open-weight models show limited and inconsistent image-dependent performance: small accuracy gains, of which only Llama-4-Scout’s on the diagnosis category survives multiplicity correction (and there stays below the most common key’s share), answers that change with the image, donor tracking whose interval excludes zero only for MedGemma-4B on sequence and Llama-4-Scout on the diagnosis category (neither separable from a random donor assignment), and no result above the best category lookup on the management rule. These results do not identify whether the models use the image internally, or why, and model scale, training, serving stack and 4-bit serving are not separated from the gap. Clinician review of the keys is the outstanding step, and only one proprietary reference model is run.

Limitations

Clinician review is outstanding. No clinician has judged whether each question is answerable

from the inputs shown or whether each key is right. The LIL and DSCR keys are computed from automated segmentation masks (DeepBraTumIA) that the model does not see, so they are targets that a reader can recover from the slices only if the segmentation is right, and segmentation errors propagate to them. The reference model’s results support the feasibility of the AIA and LIL tasks, not the validity of every item. The enhancement rule ignores T2/FLAIR change, clinical status and steroid dose, abstains where no measurable enhancing disease exists (44 of 241 follow-ups), and reproduces the expert category for 38% of the determinate ones; disagreement can come from segmentation or measurement error, from a different target lesion or reference scan, from the simplified rule, or from evidence outside the displayed slices, and the audit does not separate these. Consequently our complete response label is weaker than clinical RANO’s: it is assigned whenever follow-up enhancement falls below the measurability threshold, which can include residual sub-threshold enhancing disease, not clinical RANO’s sustained disappearance of all enhancing and nonmeasurable disease. The rule also tracks a single lesion (the enhancing component nearest the baseline target lesion’s location) and has no RANO new-lesion criterion, under which an unequivocal new lesion is itself progressive disease; of the 46 follow-ups where the rule says complete response but the expert rated progressive disease, 22 show a distinct, newly measurable enhancing component elsewhere consistent with a missed new lesion, 8 show a similarly sized component already present at the nadir, and 16 show no other measurable enhancing component at all, so this disagreement is not entirely explained by the tracking rule. The expert rating is recorded but is not the key.

The management key is a rule, not a documented treatment. LUMIERE records no treatment after chemoradiotherapy. The TCM key rests on an assumed radiotherapy schedule (ending 10 weeks after surgery) and on a simplified reading of the 12-week RANO window, which also allows progression inside the window when new enhancement lies outside the radiation field. The PJRF outcome assumes that survival time and scan weeks share the surgery-week origin, and censoring is not documented. With 45 patients a leave-one-out logistic model on the stated facts did not improve on the base rate (Brier 0.283 against 0.246), which does not show that the facts carry no predictive informa-

tion; the base rate reflects our stratified selection, and because the outcome construction is unverified PJRF is secondary and supports no claim here.

Power and scope. With 38 to 46 patients per phase, the pre-specified ± 10 -point margin is reachable only when few items differ between arms, and several cells will be inconclusive. Only one E1 image-effect test (Llama-4-Scout on DSCR) survives Holm correction, and three of the four open models (MedGemma-4B, Gemma-3-12B, Gemma-3-27B) fail the plan’s own AIA positive-control gate, so under the preregistered rule their LIL and DSCR effects support no conclusion at all, not merely a weak one. The positive control tests one skill, sequence identification; passing it does not show that a model can read lesions. The donor images come from different patients, so anatomy differs beyond the tested feature; stems in the text-only arm still refer to images. Patients are from one center and the cohort was stratified by design; the four open-weight models run at 4-bit through Ollama while the control is an API model of unknown size and precision, so the gap between them mixes scale, quantization and training and cannot be attributed to shortcut learning; a bf16 rerun of Gemma-3-27B (through vLLM, so precision is confounded with the serving stack) stays near chance on AIA, which does not resolve the gap but cannot isolate quantization from the serving stack, and no other model was rerun. There is one reference model, run once at temperature 0, and API models change without notice. **Input handling was checked only in part.** We confirmed post hoc, for each pipeline, that every image of a question raises the prompt token count (including the second and third DSCR images) and that known-answer probes are answered correctly (Appendix F). We did not compare the images each serving stack passes to its encoder with a reference implementation, and a check on one patient cannot show that MRI contrast is preserved by each model’s image encoder. Near-chance sequence identification in several open models therefore remains unexplained: the check makes a dropped-image bug unlikely but does not exclude weak encoder features or default-answer behavior. MedGemma-4B returned invalid JSON for 105 of 965 records, and Llama-4-Scout for 46 of 965; the answer letter was recovered by pattern for all but 0 records (Appendix F). The comparison to the reference model mixes scale, training, quantization and serving. Intervals treat patients as exchangeable.

Ethics Statement

This work evaluates the diagnostic reasoning of general-purpose and medical-domain language models on real, de-identified patient imaging data; it does not deploy, recommend, or endorse any model for clinical use, and no results in this paper should be interpreted as a clinical validation of any evaluated system. LUMIERE is used under its published non-commercial research license, and all patient data is accessed in its already de-identified, publicly released form; no new identifiable patient information is collected. The LUMIERE authors report that the cantonal ethics committee of Bern approved the studies and waived written informed consent, that the images are skull-stripped, and that dates are given as weeks relative to the pre-operative scan; we did not attempt to re-identify patients. An AI coding assistant (Claude Code) helped write and run the analysis code and edit the text; the authors are responsible for all content.

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		A OmniBrainBench Compatibility Details	
		A chain presupposes that a case’s phases are one patient’s trajectory. In the 6,823-question closed-ended subset of OmniBrainBench (Peng et al., 2026) only 2 verifiable cross-phase patient links exist (both in VQA_RAD, one AIA and one LIL question); stripping modality, view and slice suffixes from filenames gave the same result. The release has no complete patient identifier, so these are	1000
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lower bounds. Each sub-corpus was built for one clinical task, so a patient is essentially asked one phase-type of question, and reading `source_file` as patient identity would stitch together different patients. This is an incompatibility with our evaluation, not a defect in the benchmark. The supplement’s patient-level ADNI extension is not a brain-tumor cohort and, as of September 2026, has label-generation scripts only; we did not evaluate it. We found no dataset with same-patient, multi-phase continuity and a question layer (Section 2), which motivates LUMIERE.

B Artifacts, Licenses, Compute and Intended Use

Data. LUMIERE (Suter et al., 2022) (91 patients, from Figshare) may be used only for non-commercial purposes, which our research use respects; its authors report ethics approval and a consent waiver by the cantonal ethics committee of Bern, skull-stripped images and dates blurred to weeks. We share the rendered slices the models saw, with attribution and for non-commercial research only, not the imaging archive. OmniBrain-Bench (CC BY-SA 3.0 per its dataset card) was used only to count cross-phase links (Appendix A) and is not redistributed.

Reproducibility artifact. An anonymous repository (link in the submission) holds the items with keys and key origins, the donor assignments, the exact prompt of every single-question condition, every model response used, a manifest of which run supplied each record and which later records replaced earlier ones, the analysis code with fixed seeds, and one command that regenerates every statistic and table and checks them against this paper. It excludes model weights, API keys and the full LUMIERE archive.

Models and terms. MedGemma-4B (Health AI Developer Foundations terms), Gemma-3-12B and Gemma-3-27B (Gemma Terms of Use), Llama-4-Scout (Llama 4 Community License) and Gemini-3.6-Flash (Google’s API terms) were used for research evaluation only, as these terms permit; no output is used clinically. Tags, digests and quantization are in Table 8. The MedGemma and Gemma 3 technical reports are cited only to identify those models; they are not peer-reviewed, and no claim rests on them. Llama-4-Scout has no technical report.

Software and compute. Ollama 0.33.0 served 4-bit Q4_K_M builds of the four open models (4.3, 12.2, 27.4 and 108.6 billion parameters; Llama-4-Scout is a mixture of experts with 17B active) and vLLM 0.27.1 (PyTorch 2.13.0, Transformers 5.15.1) the bf16 run; the analysis used Python 3.12.5 with NumPy 2.3.5, SciPy 1.18.1, pandas 3.0.5 and Pillow 12.3.0. Almost all GPU time was on one NVIDIA RTX PRO 6000 Blackwell GPU per job: about 154 GPU-hours for all LUMIERE work including failed and exploratory jobs, about 56 for the runs and input checks reported here. The Gemini runs made 525 API calls and about 1.1 million tokens including hidden reasoning tokens. Decoding was greedy with the caps in Section 4; no hyperparameters were tuned and each condition ran once.

Intended use. The items, code and results are for evaluation research only: not a clinical benchmark, keys not clinician-validated, and nothing here supports using a model for diagnosis, prognosis or treatment.

C A Worked Example

The patient is Patient-007, the first in id order with all five phases and a three-image DSCR item (a rule fixed before any answer was inspected). Figure 1 shows the images the models see, and the donor images used in the swap.

AIA (positive control). The image is a single axial slice from a brain MRI examination of a patient with glioblastoma (skull-stripped and registered to a standard template). Which MRI sequence does this image show? *Options as shown:* A) FLAIR (fluid-attenuated inversion recovery); B) T2-weighted; C) T1-weighted, before contrast; D) T1-weighted, after gadolinium contrast. *Key:* D) T1-weighted, after gadolinium contrast.

LIL. The image is an axial slice of a post-contrast T1-weighted brain MRI in radiological convention (the patient’s right side is on the left of the image; anterior is at the top). Where is the main tumor region (enhancing tumor, necrosis or resection cavity, not the surrounding edema) on this slice? Divide the brain on this slice into right and left hemispheres at the midline and into anterior and posterior halves at the midpoint of its front-to-back extent. *Options as shown:* A) Right hemisphere, posterior half; B) Left hemisphere, anterior half;

C) Right hemisphere, anterior half; D) Left hemisphere, posterior half. *Key:* A) Right hemisphere, posterior half.

DSCR. Three post-contrast T1-weighted axial slices of the same patient are shown, each at the largest cross-section of the same enhancing lesion (the enhancing lesion nearest the largest enhancing region on the post-operative baseline scan, followed across scans), in time order: the post-operative baseline scan (image 1), the earlier scan with the smallest enhancing lesion (the nadir, image 2) and the follow-up scan (image 3). Pixels are 1 mm in the original scan, and a 20 mm scale bar is drawn on each image. Apply the RANO enhancement criteria to the enhancing lesion. Measurable means at least 10 mm by 10 mm. Progression is a 25% or greater increase in the product of the two largest perpendicular diameters relative to the nadir, or a new measurable enhancing lesion. Response is judged against the post-operative baseline: partial response is a 50% or greater decrease, complete response is no remaining measurable enhancing disease, otherwise stable disease. T2/FLAIR changes, steroid dose and clinical status are not provided. What is the response category at follow-up? *Options as shown:* A) Partial response; B) Progressive disease; C) Complete response; D) Stable disease. *Key:* B) Progressive disease.

The key is computed from the bidimensional products of the largest enhancing component on the three displayed slices: 10 mm² at the post-operative baseline, 0 mm² at the nadir and 385 mm² at follow-up; there is no measurable enhancing disease at the nadir and a measurable lesion at follow-up (a new measurable lesion), which the stated rule calls progressive disease. The expert rating for this follow-up is PD (rationale: “Target L.: 1, 16mm x 19mm”), the same category as the key, so it is not one of the disagreements counted by the audit. The rating is recorded for every item and is never the key.

TCM. A 47-year-old male with glioblastoma (MGMT status not available; IDH status not available) had a complete resection of the enhancing tumor and then standard chemoradiotherapy (radiotherapy with concurrent temozolomide), followed by adjuvant temozolomide. No second surgery or second-line treatment has been given. This MRI is 63 weeks after surgery; chemoradiotherapy ended about 53 weeks before this scan. Using your earlier assessment of this patient, what is the most appropriate next step? *Options as shown:* A) Continue

adjuvant temozolomide unchanged and plan the next surveillance MRI in 8 to 12 weeks; B) Stop tumor-directed treatment and move to best supportive care with no further imaging plan; C) Change therapy now: bring re-resection, second-line treatment or a clinical trial to tumor board; D) Continue adjuvant temozolomide for now and repeat the MRI in 4 to 8 weeks before changing therapy. *Key:* C) Change therapy now: bring re-resection, second-line treatment or a clinical trial to tumor board. The rule is: non-PD → continue; PD and ≤ 12 weeks after radiotherapy → repeat MRI first; PD and later → change therapy (RANO 2010; Stupp schedule assumed). Here the category is progressive disease and chemoradiotherapy ended 53 weeks before the scan, so the class is escalate. Changing the stated timing across the 12-week window, or replacing the stated category, changes the rule’s answer (the fact flips of E3).

PJRF (secondary). A 47-year-old male with glioblastoma (MGMT status not available; IDH status not available) had a complete resection of the enhancing tumor and then standard chemoradiotherapy (radiotherapy with concurrent temozolomide), followed by adjuvant temozolomide. No second surgery or second-line treatment has been given. The follow-up MRI is 63 weeks after surgery and the patient is alive at this time. Using only the information above and your earlier assessment of this patient, estimate the probability that the patient will die within 52 weeks after this scan. The recorded outcome is death within 52 weeks; the forecast is scored by the Brier score.

Donor swap. For the swap condition the AIA image is replaced by the one from Patient-054 (key FLAIR (fluid-attenuated inversion recovery) instead of T1-weighted, after gadolinium contrast), the LIL image by the one from Patient-023 (key Left hemisphere, posterior half instead of Right hemisphere, posterior half), and the DSCR images by those of Patient-019 (key Partial response instead of Progressive disease); the stem and options are unchanged, so the donor’s answer is always an option.

D Item Set Summary and Donor-Dependence Checks

Item set. From 59 patients with a first-line follow-up we keep 46. AIA has 46 items (T1 12, T1 after contrast 12, T2 11, FLAIR 11). LIL has 38:

	AIA	LIL	DSCR
Key / donor key	T1c / FLAIR	R post / L post	PD / PR
Gemini-3.6-Flash (reference)	T1c / T1c / FLAIR	R post / R ant / L post	PD / PD / CR
Llama-4-Scout	T1c / T1c / T1	L post / L post / L post	PD / SD / PD
Gemma-3-27B	T2 / T1c / FLAIR	L ant / L ant / L post	PD / PD / PD
Gemma-3-12B	T2 / T2 / FLAIR	R ant / R ant / L post	PD / PD / PD
MedGemma-4B	T1 / T1 / T1	R ant / R ant / L ant	CR / CR / CR

Table 5: Answers for Patient-007 under the own-image / text-only / swapped-image conditions (same order in every cell), and the key and donor key. Labels: T1c is T1 after contrast; L/R hemisphere, ant/post half; PD, SD, PR, CR are the RANO categories.

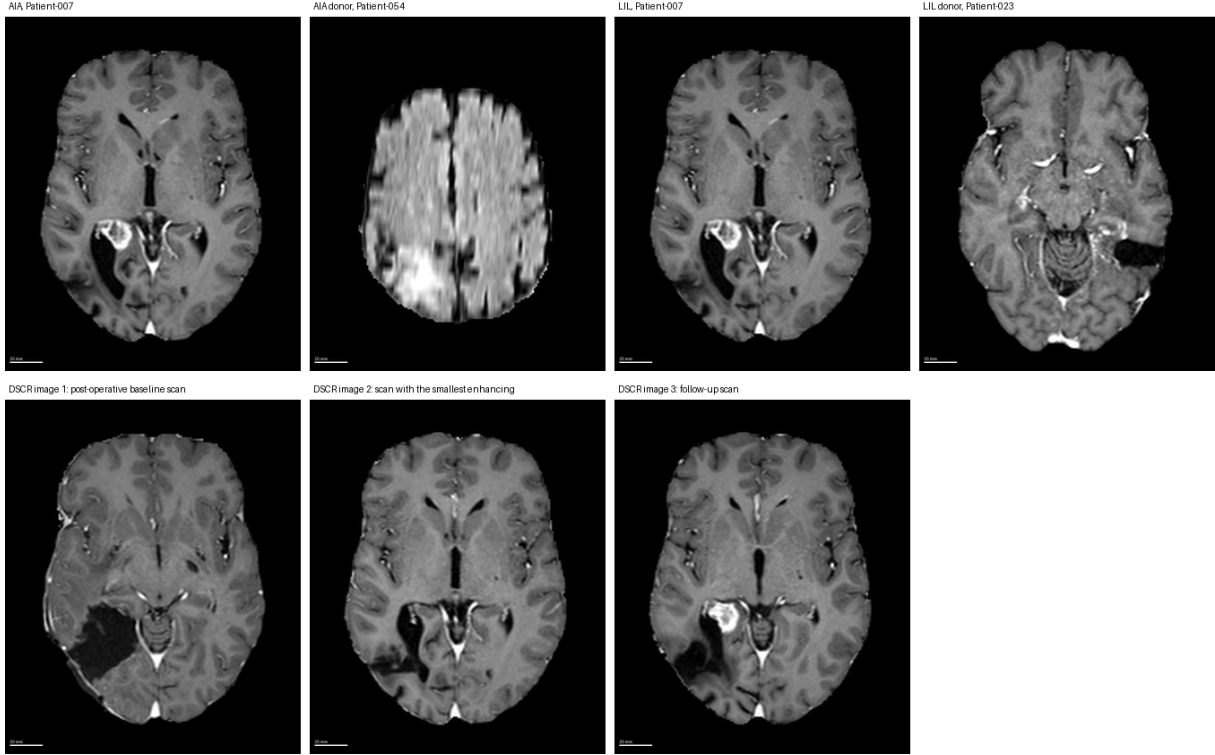


Figure 1: Images for Patient-007. Top: the AIA slice and its donor, the LIL slice and its donor. Bottom: the three DSCR slices (baseline, nadir, follow-up). Slices are atlas-space, skull-stripped, magnified $4\times$, with a 20 mm scale bar.

the rest have no tumor core, a core within 10 mm of a midline or crossing it, or one under 100 mm^2 (posterior left 11, posterior right 11, anterior right 9, anterior left 7). DSCR has 46 (progressive disease 27, complete response 9, stable disease 6, partial response 4), the key equalling the expert rating for 26. TCM has 46 (continue 19, repeat MRI first 7, change therapy 20). PJRF has 45: a patient needs a recorded survival time after the scan week. A swap needs a donor with a different key and the same number of images; each patient gets one donor per phase and donates to at most two recipients, leaving 46 AIA, 38 LIL and 42 DSCR pairs (four patients have no compatible DSCR donor).

Audit size check. Where the rater recorded target-lesion diameters (81 follow-ups, 69 of them rated progressive disease), agreement was the same whether our product was within a factor of 1.5 of the rater’s (21 of 33) or not (22 of 48), so size discrepancy does not explain the disagreements in that subset; it has few stable-disease or partial-response ratings, where disagreement concentrates, so it cannot test them.

Donor dependence (post hoc). Each phase uses 31 to 30 distinct donors, none used more than 2 times, and every patient is both donor and recipient, so a component bootstrap is uninformative. We instead resample recipients that share a donor to-

gether, drop each donor in turn, and permute donor keys among recipients (never onto a recipient’s own key; 5,000 permutations; p is the share with a gain at least as large). MedGemma-4B on AIA: gain +10.9, donor-cluster interval [+2.1, +22.7], leave-one-donor-out +6.8 to +11.4, permutation $p = .059$. Llama-4-Scout on DSCR: gain +28.6, interval [+2.5, +53.9], leave-one-donor-out +25.0 to +35.0, permutation $p = .244$. No permutation p of the twelve open-model cells is below .05.

E Donor-Swap and Reference-Model Tables

Tables 6 and 7 give the donor-swap results for the open models and the reference-model results.

F Output Handling, Input Check and Fact-Flip Transitions

These are post hoc, descriptive checks. Invalid JSON (answer letter recovered by pattern matching, since the prompt orders it first) and records with no recoverable letter, of 965 records each (386 for the reference and precision-control models): MedGemma-4B 105 invalid, 0 no-letter; Llama-4-Scout 46 invalid, 0 no-letter; Gemma-3-27B 1 invalid, 0 no-letter; Gemma-3-12B, Gemini-3.6-Flash and Gemma-3-27B-bf16 all 0 and 0. MedGemma-4B’s invalid JSON is truncation at the 800-token cap after repeating one sentence, as is Gemma-3-27B’s single case; Llama-4-Scout’s is a misspelled field name (a space inside visual_grounding) in an otherwise complete response. The letter is scored in all. MedGemma-4B’s DSCR accuracy on valid-JSON records (23.1%) is close to all records (19.6%), so parsing does not explain its below-chance score, which comes from the letters it chooses (never the fourth option, in either arm). These counts do not show that images reach each model; the input check does.

Input check. For each pipeline we sent one real AIA, one LIL and one three-image DSCR item with zero to three of its images behind a unique nonce (preventing prefix caching) and read the prompt token count. Each added image raises it by a similar amount (Table 8), so none was dropped, including the second and third DSCR images. The same pipelines passed known-answer probes (solid colours, printed numbers, image order and count, a slice’s black background), and one slice stayed legible squashed to 896 and 448 pixels. Run once on one patient, this shows images reach the model and

simple stimuli are perceived, not that MRI contrast survives each image encoder.

Table 9 gives each item’s transition under a fact flip relative to the same item under the true context: moved to the rule’s new answer, already there, unchanged elsewhere, or changed to another answer. Only the first column shows an effect of the flip. The last column is the registered summary (moved plus already there, over all items); on the label flip it is highest for MedGemma-4B, largely because 21 of its answers were already the rule’s new answer.

PJRF outcome construction. We assume the survival time shares its origin with the week numbering; LUMIERE does not state the origin of overall survival or mention censoring. 4 of 86 patients with a survival time have one shorter than their last imaging week (1 in our cohort), and 4 of the 45 outcome labels would change under a four-week shift of the origin. A leave-one-out logistic model on age, MGMT, resection extent and weeks since surgery reaches Brier 0.283 against 0.246 for the constant base rate (AUC 0.387), so it did not beat the base rate in this small sample, which does not show that the facts carry no information.

PJRF (E5, secondary). No model’s Brier score differs clearly from the constant base rate: the paired difference against the leave-one-out base rate (patient-bootstrap 95% interval, negative is better) is -0.013 for MedGemma-4B, $+0.001$ for Gemma-3-12B, -0.017 for Gemma-3-27B and $+0.009$ for Llama-4-Scout. MedGemma-4B and Gemma-3-12B choose the 0.5 bin for 78% and 91% of patients, so their Brier score is close to 0.25 by construction. Because the outcome’s time origin and censoring are unverified (Section 3.2), we draw no conclusion from PJRF.

Model	Phase	<i>n</i>	Own	Swap: donor	Text: donor	Gain [95% CI]	Add-half CI	Exact <i>p</i>	Swap changes
MedGemma-4B	AIA	46	34.8	32.6	21.7	+10.9 [+2.2, +19.6]	[+0.9, +20.0]	.062	32.6
MedGemma-4B	LIL	38	23.7	21.1	28.9	−7.9 [−18.4, +2.6]	[−19.3, +4.3]	.375	13.2
MedGemma-4B	DSCR	42	21.4	28.6	23.8	+4.8 [−4.8, +14.3]	[−5.3, +14.4]	.625	9.5
Gemma-3-12B	AIA	46	26.1	17.4	17.4	+0.0 [−13.0, +13.0]	[−12.2, +12.2]	1.00	50.0
Gemma-3-12B	LIL	38	31.6	23.7	18.4	+5.3 [−10.5, +18.4]	[−9.6, +19.6]	.727	52.6
Gemma-3-12B	DSCR	42	54.8	35.7	31.0	+4.8 [−4.8, +14.3]	[−5.3, +14.4]	.625	2.4
Gemma-3-27B	AIA	46	32.6	34.8	30.4	+4.3 [−4.3, +13.0]	[−4.9, +13.2]	.625	32.6
Gemma-3-27B	LIL	38	23.7	26.3	21.1	+5.3 [−7.9, +18.4]	[−7.9, +17.9]	.688	44.7
Gemma-3-27B	DSCR	42	54.8	35.7	33.3	+2.4 [−4.8, +9.5]	[−6.6, +11.2]	1.00	7.1
Llama-4-Scout	AIA	46	37.0	30.4	28.3	+2.2 [−6.5, +13.0]	[−7.9, +12.1]	1.00	32.6
Llama-4-Scout	LIL	38	39.5	31.6	23.7	+7.9 [−7.9, +23.7]	[−7.8, +22.8]	.508	55.3
Llama-4-Scout	DSCR	42	54.8	50.0	21.4	+28.6 [+7.1, +50.0]	[+5.6, +49.0]	.029	47.6

Table 6: E2: share of answers (%) that equal the donor’s key when the image is swapped, and when there is no image; the gain is the difference in percentage points, with its bootstrap interval, an add-half sparse-data interval (Add-half CI) and the exact McNemar *p* (unadjusted); the last column is the share of items whose answer differs from the own-image answer after the swap.

Phase	<i>n</i>	Image	Text-only	Δ [95% CI]	Swap: donor key	Gain [95% CI]
AIA	46	95.7	21.7	+73.9 [+58.7, +87.0]	95.7	+67.4 [+54.3, +80.4]
LIL	38	100.0	36.8	+63.2 [+47.4, +78.9]	100.0	+86.8 [+76.3, +97.4]
DSCR	46	71.7	58.7	+13.0 [+0.0, +26.1]	50.0	+16.7 [+2.4, +31.0]

Table 7: Reference model (Gemini-3.6-Flash): accuracy (%) with the image and with text only, the paired difference, the share of swapped-image answers equal to the donor’s key, and the donor-tracking gain (percentage points).

Model	Tag	Digest	Quantization	Tokens per image	Probes passed
MedGemma-4B	medgemma:4b	9fe4e9a6c9bd	Q4_K_M	285–308	8/8
Gemma-3-12B	gemma3:12b	f4031aab637d	Q4_K_M	284–305	8/8
Gemma-3-27B	gemma3:27b	a418f5838eaf	Q4_K_M	280–304	8/8
Llama-4-Scout	llama4:scout	bf31604e25c2	Q4_K_M	1456–1477	8/8
Gemma-3-27B-bf16	google/gemma-3-27b-it	–	bf16 (vLLM)	271–288	8/8

Table 8: Input check per pipeline: model tag, digest and quantization as reported by the server, the range of prompt-token increments per added image, and known-answer probes passed out of eight.

Model	Flip	Items	Moved	Already there	Unchanged	Other	Moved / informative	Registered share (%)
MedGemma-4B	label (all)	46	10	21	12	3	10/25	67.4
MedGemma-4B	timing (PD)	27	1	0	23	3	1/27	3.7
Gemma-3-12B	label (all)	46	18	6	10	12	18/40	52.2
Gemma-3-12B	timing (PD)	27	8	6	11	2	8/21	51.9
Gemma-3-27B	label (all)	46	25	0	2	19	25/46	54.3
Gemma-3-27B	timing (PD)	27	0	7	20	0	0/20	25.9
Llama-4-Scout	label (all)	46	10	9	15	12	10/37	41.3
Llama-4-Scout	timing (PD)	27	11	6	10	0	11/21	63.0

Table 9: TCM fact flips: transitions relative to the true-context answer for the same item. The label flip covers all 46 items; the timing flip covers the 27 progressive-disease items.